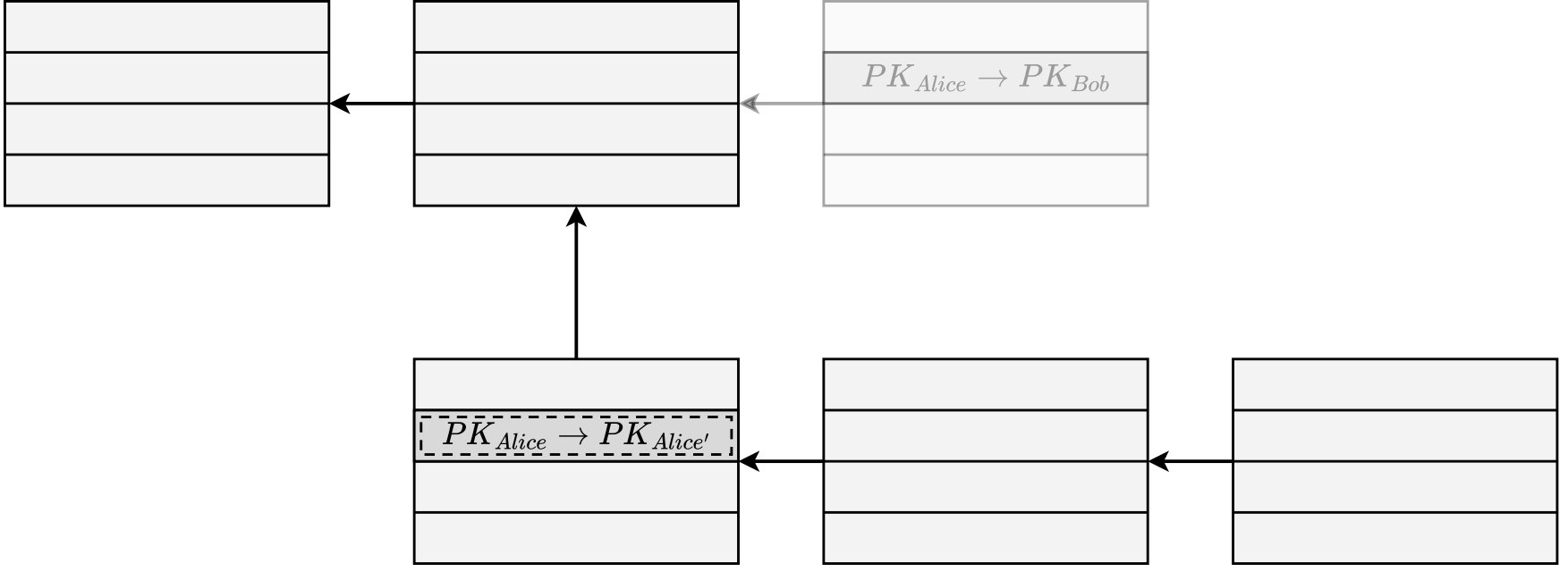
$PK_{Alice} \rightarrow PK_{Bob}$
$PK_{Alice} \rightarrow PK_{Alice'}$